FORENSIQ: AI-POWERED DIGITAL FORENSICS

Comprehensive Technical Reference, Architecture, Baselines & Operational Handbook

Framework: Autonomous Digital Forensics & Incident Response (DFIR) with Machine Learning & XAI
Canonical Pipeline: 12-Stage Automated Triage Engine | 17-Feature Forensic Space
Evaluated Evidence: NIST CFReDS Data Leakage Scenario (Windows 7 SP1 x64, 85,017 Files)
Compliance: NIST SP 800-86 / ISO/IEC 27042 / MITRE ATT&CK; v14 Standards

1. System Architecture: The 12-Stage ForensIQ Pipeline

ForensIQ orchestrates end-to-end evidence ingestion, artifact parsing, feature engineering, unsupervised anomaly scoring, temporal burst detection, confidence tiering, ATT&CK; mapping, Explainable AI attributions, interactive timeline synthesis, and scientific baseline benchmarking:

Pipeline Stage & Core Script	Forensic Operation & Technical Deliverables
Stage 1: Evidence Ingestion & Partition Probing extractor.py	Parses raw disk images (.raw/.dd) and E01 files without partition mounting. Extracts MFT filesystem metadata, EVT/EVTX logs, Prefetch, and Registry hives.
Stage 2: Cryptographic Manifest Verification chain_of_custody.py	Computes streaming dual hashes (SHA-256 and MD5) across all 102 extracted artifacts, establishing a verifiable baseline manifest.
Stage 3: 17-Feature Forensic Engineering feature_engineering.py	Extracts 17 numerical features per file: MACB deltas with zero-timestamp masking, size, deleted bitmask flags, executable/archive extensions, user/temp paths, and prefetch execution indicators.
Stage 4: Isolation Forest Anomaly Detection (v1) anomaly_detection.py	Trains Isolation Forest on the standardized 17-feature matrix. Flags the top anomalous records into flagged_anomalies.csv and bundles (model, scaler, cols).
Stage 5: Sliding-Window Temporal Burst Detection (v2) burst_detection.py	Performs moving-window aggregation across MFT modification and EVTX event timestamps to detect rapid staging, wiping, or encryption bursts.
Stage 6: Multi-Artifact Confidence Calibration confidence_tiers.py	Classifies anomalous candidates into CRITICAL, HIGH, MEDIUM, and LOW risk tiers based on decision boundary quartiles.
Stage 7: MITRE ATT&CK; Technique Mapping mitre_mapper.py	Maps anomaly signatures to MITRE ATT&CK; Enterprise techniques (T1070.006 Timestomping, T1052.001 USB Exfiltration, T1036.005 Masquerading).
Stage 8: Explainable AI Attribution (TreeSHAP) xai_shap.py	Computes game-theoretic Shapley attributions (TreeExplainer) on raw feature data (data=X_flagged_raw), generating individual waterfall charts and global beeswarm plots.
Stage 9: Unified Chronological Timeline timeline.py	Fuses MFT filesystem events and EVTX security logs into a timezone-normalized UTC timeline with client-side search and 50-event pagination.
Stage 10: Automated Incident Remediation remediation.py	Synthesizes actionable containment PowerShell scripts tailored to the detected threat class (account lockout, timestamp reset, malware quarantine).
Stage 11: Master Report & Audit Assembly forensiq_report.py	Compiles executive breach classification, CoC hash verification, verifiable ground-truth scoring, and an interactive HTML dashboard.
Stage 12: Scientific Baseline Benchmark evaluate_baselines.py	Executes equal-budget comparisons (K=25, 50, 100, 250, 500, 1000, 4250) against Random, Heuristic Rules, Univariate, Isolation Forest, and k-NN.

2. The Canonical 17-Feature Forensic Feature Space

Every feature possesses a rigorous forensic definition, unit, and calculation formula, designed to differentiate malicious actions from benign OS operations:

Feature	Unit / Source	Exact Formula	Forensic Purpose
delta_mc	Seconds / MFT	|m_time - c_time| if both > 0 else 0	Detects timestomping metadata manipulation.
delta_mb	Seconds / MFT	|m_time - cr_time| if both > 0 else 0	Detects file backdating prior to creation.
delta_ab	Seconds / MFT	|a_time - cr_time| if both > 0 else 0	Detects anomalous access vs birth latency.
future_m	Binary / MFT	1 if m_time > now_ts else 0	Flags timestamps set into the future.
zero_size_recent	Binary / MFT	1 if size==0 and age < 24h else 0	Flags zero-byte touch/wiper artifacts.
size	Bytes / MFT	meta.size	Detects massive data archives or tiny binaries.
is_deleted	Binary / MFT	bool(flags & UNALLOC)	Flags shift-deleted records in free space.
is_exe	Binary / Path	.exe, .dll, .sys, .scr	Isolates executable machine code.
is_archive	Binary / Path	.zip, .rar, .7z, .tar, .gz	Flags data compression staging files.
temp_path	Binary / Path	contains /temp/ or /tmp/	Detects malware staging in temporary folders.
user_path	Binary / Path	contains /users/ or /docs & settings/	Separates user activity from core OS activity.
credential_tool	Binary / Prefetch	mimikatz, pwdump, cain, 123wasp	Flags credential-dumping utility execution.
sniffer_tool	Binary / Prefetch	wireshark, ethereal, tcpdump, winpcap	Flags network packet sniffing execution.
powershell_exec	Binary / Prefetch	powershell.exe in prefetch	Flags script-based living-off-the-land attacks.
cmd_exec	Binary / Prefetch	cmd.exe in prefetch	Flags command shell execution for manual staging.
script_host	Binary / Prefetch	wscript or cscript in prefetch	Flags Windows Script Host execution.
usb_present	Binary / Registry	SYSTEM USBSTOR non-empty	Corroborates physical external storage presence.

3. Scientific Evaluation & Baseline Benchmark (Milestone 5)

To establish whether machine learning adds measurable value compared to simpler approaches, ForensIQ evaluates 11 ranking methods across 7 matched investigation budgets (K) on the 85,017 files of NIST CFReDS:

Ranking Method	K=25	K=50	K=100	K=250	K=500	K=1000	K=4250 (5%)
Random Baseline (10 seeds)	0.0	0.0	0.0	0.0	0.0	0.1	1.3±1.2
Rule: .exe under /Program Files	0/19	0/19	0/19	0/19	0/19	1/19	1/19
Rule: Suspicious Paths (User/Temp)	0/19	0/19	1/19	**12/19**	**12/19**	**12/19**	**13/19**
Univariate: delta_mc	0/19	0/19	0/19	0/19	0/19	0/19	0/19
Univariate: File Size	0/19	0/19	0/19	0/19	0/19	1/19	6/19
Isolation Forest (17 Feats)	0/19	0/19	0/19	0/19	0/19	0/19	**7/19**
Isolation Forest Multi-Seed (0–9)	0.0	0.0	0.0	0.0	0.0	0.0	6.6±1.6
k-NN Distance Outlier (k=5)	0/19	0/19	0/19	0/19	3/19	4/19	**12/19**
Ablation: MACB Deltas Only	0/19	0/19	0/19	0/19	0/19	0/19	0/19
Ablation: Path Heuristics Only	0/19	0/19	0/19	0/19	0/19	0/19	2/19

4. MITRE ATT&CK; Evidence Cards & Corroboration Protocols

In accordance with forensic science standards, ForensIQ replaces aggressive heuristic leaps with evidence-linked observation cards that identify required corroborating artifacts:

ATT&CK; Technique	Observed Forensic Signal	Alternative Explanations	Required Corroboration	Scientific Status
T1070.006 Timestomping	Divergence between $SI Modified and Record Changed times (delta_mc > 86400s).	Archive extraction; software installers unpacking older legacy DLLs.	Compare $STANDARD_INFORMATION ($SI) with kernel-protected $FILE_NAME ($FN) timestamps and USN Journal.	**HYPOTHESIS / SUSPICIOUS DELTA**
T1070.001 Log Clearing	Hourly continuity gap in event logs during business hours (event_count == 0).	Workstation powered off, suspended, or disconnected from network.	Event ID 1102 (Security cleared) or Event ID 104 (System cleared); power state uptime verification.	**OBSERVED GAP / NEEDS POWER CORROB.**
T1052.001 USB Exfiltration	Registry key SYSTEM\CurrentControlSet\Enum\USBSTOR contains connected drive.	Authorized business thumb drive; connection without data copy.	LNK shortcut files or Shellbags proving navigation of removable drive letter; MFT unallocated file copy traces.	**POSSIBLE CONNECTION / REQUIRES FILE-COPY**
T1486 Ransomware	Multiple archive files (.zip, .rar, .7z) flagged in user folders.	Normal user compression; backup archives; downloaded installers.	Presence of ransom notes; Shannon entropy > 7.8 across document extensions; dropped locker binaries.	**UNVERIFIED / DATA COMPRESSION ONLY**
T1003.002 SAM Dumping	Prefetch execution record matching pwdump, cain, or mimikatz.	Antivirus scanning SAM hive; Volume Shadow Copy (VSS) routine backups.	Prefetch execution count > 0; Event ID 4688 process execution logs; output hash dump text files.	**HIGH RISK / PENDING RUNTIME PROOF**
T1036.005 Masquerading	File extension conflicts with binary magic header (e.g. .jpg containing Excel XML).	Software bug or inadvertent user file extension naming.	Magic byte file header confirmation; LNK shortcuts showing file opened in conflicting host application.	**CONFIRMED MASQUERADING**